OPTIQ

NETRASENSE

NetraSense helps early stage detection of diabetic retinopathy in rural areas using AI detection, quality checks and real-time clinician feedback to instantly grade scans and detect retinal lesions.

DEVADEEP MISRA
DRIPTO MAITY
SHUVRO GUPTA

OPTIQ

OVERCOMING FRONTLINE SCREENING BARRIERS

PROBLEM AND TARGET USERS

WHAT SPECIFIC PROBLEM ARE WE SOLVING?

Unreadable fundus captures and poor lighting cause false AI diagnoses and missed diabetic retinopathy.

WHO FACES THE PROBLEM?

Rural households which require retinal screening of people there and overburdened retinal specialists.

HOW DO THEY HANDLE IT CURRENTLY, AND WHAT IS MISSING?

Current Handling: Manual specialist review and rigid commercial AI.

What's Missing:

- No recapture feedback for bad photos
- Black-box grades without lesion maps
- Zero ability to correct AI mistakes online.

WHAT NEEDS TO HAPPEN FOR REAL-WORLD USE?

Prospective multi-center trials across diverse patient populations and hardware devices, alongside regulatory clearance, clinical oversight, structured training for community health workers, and seamless referral integration.

SOLUTION

WHAT WE BUILT

OPTIQ

SOLUTION, FEATURES AND SYSTEM FLOW

Input: 3D-printed smartphone fundus capture. Action: MATLAB quality checks verify usability before analysis. Result: Instant ICDR grade, lesion contours, & RLHF doctor feedback loop.

TECHNOLOGIES AND INTEGRATIONS

- Hardware & App: 3D-printed camera prototype & Flutter mobile app.
- Backend & AI: Python backend, PyTorch models, MATLAB image enhancement, & Agent-37 reporting.

BUILT DURING BUILD 2026

- Reinforcement Learning through Human Feedback: Clinician click feedback loop for model weight adaptation.
- Quality Metrics: MATLAB-based extraction for accepting/rejecting fundus scans.
- App Backend: Full backend development linking Flutter app to Python.
- Hardware: 3D modeling & partial prototype assembly.
- Integration: Agent-37 automated clinical summary generation.

WORKFLOW

1 Capture Fundus Image

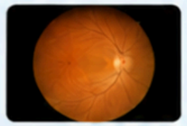




Image from fundus camera (with smartphone)

2 Extract Quality Metrics (MATLAB)

- Image clarity
- Illumination
- Focus / sharpness
- Other quality metrics

MATLAB

3 Image Usability Check (ML Model)

Uses quality metrics to determine if the image is usable.

ML Model

Image usable?

No

4 Recapture Image

Request user to capture another image.

Yes

5 Image Enhancement (MATLAB)

- Noise reduction
- Contrast improvement
- Illumination correction

MATLAB

6 AI Analysis (PyTorch)

- Lesion detection
- DR grading
- Confidence score

PyTorch

7 Doctor Reinforcement Learning

Doctor reviews and provides feedback to improve the model (over time).

Human-in-the-loop

8 Final Report (ML)

ML model generates structured results (grade, lesions, confidence, etc.).

ML Model

9 Agent37 – Generate Summary

Converts results into a clear, readable and summarised report.

Agent37 / Hermes

10 App Displays Report

Flutter app shows the final report to the user.

Flutter

Technologies Used

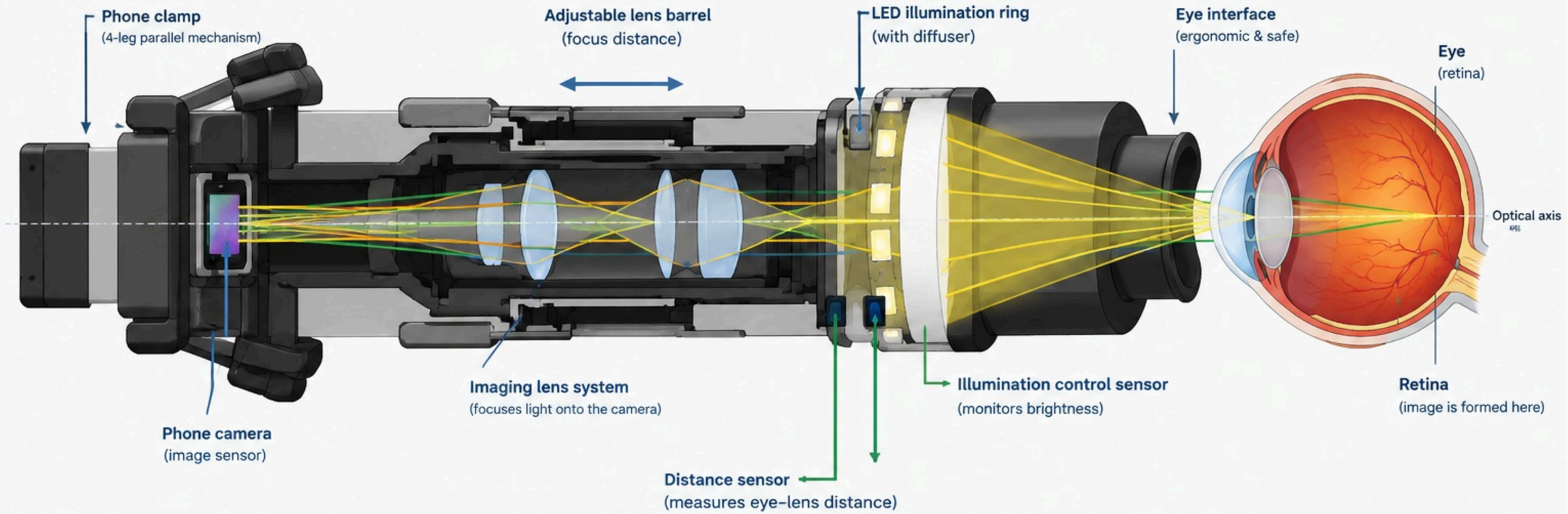
MATLAB	PyTorch	Python + FastAPI	Agent37 / Hermes	Flutter	IDRiD Dataset
(Image processing)	(AI/ML models)	(backend & integration)	(report generation)	(mobile app)	(model training)

OPTIQ

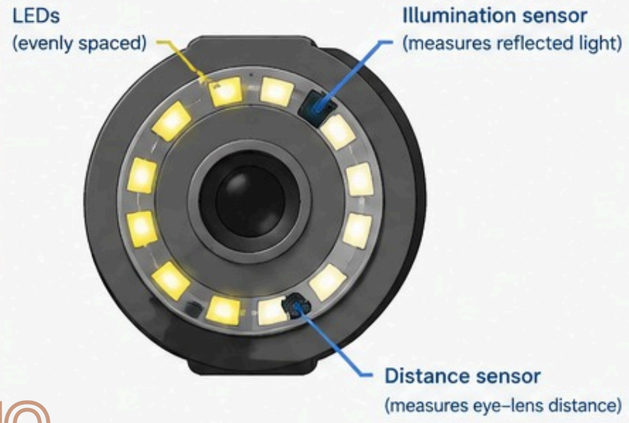
CLAMP SYSTEM

Optical Path and Sensor Integration

How light travels, how the image is formed, and how the system adjusts for the best result.



Top View – Illumination Ring & Sensors



Step-by-Step Light Path

1. LED ring emits uniform light through the diffuser, illuminating the retina.
2. Light reflects from the retina and travels back through the imaging lens system.
3. The lens focuses the reflected light onto the phone camera sensor.
4. The phone's camera captures the image and processes it.
5. Distance and illumination sensors monitor eye–lens distance and light level for optimal focus and exposure.

Key Components & Functions

- **LED illumination ring**
  Provides uniform, controlled light.
- **Illumination sensor**
  Checks light intensity for consistent exposure.
- **Distance sensor**
  Measures eye–lens distance for focus adjustment.
- **Adjustable lens barrel**
  Moves to keep the image in focus.
- **Phone camera**
  Captures the retinal image.

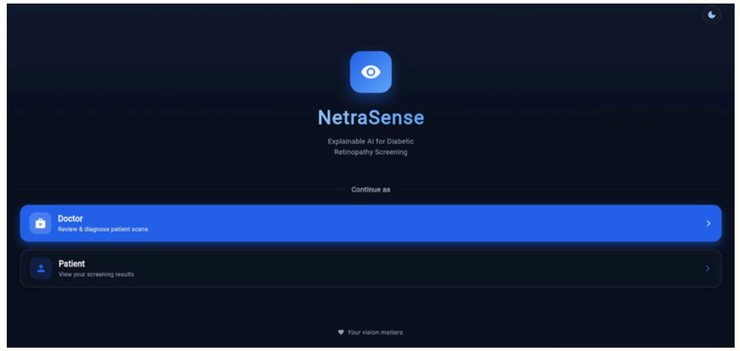
Ray Legend

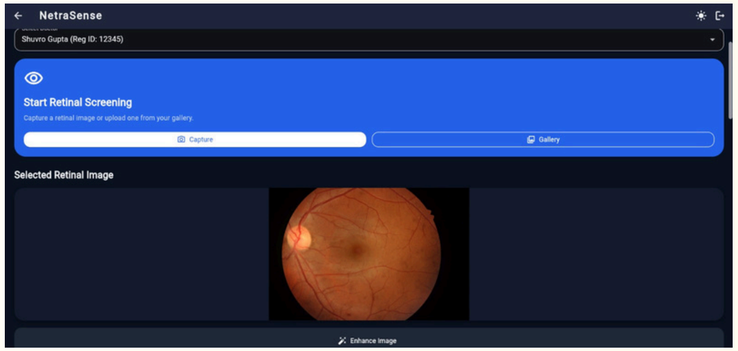
- Illumination light (from LED ring to eye)
- Reflected light (from retina to camera)
- Optical axis (center line)
- Lens element (air/glass)

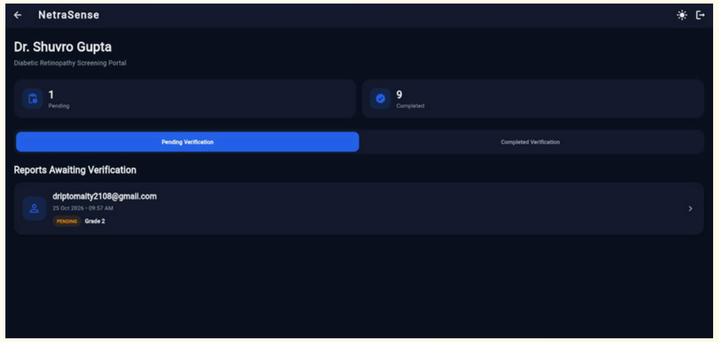
OPTIQ

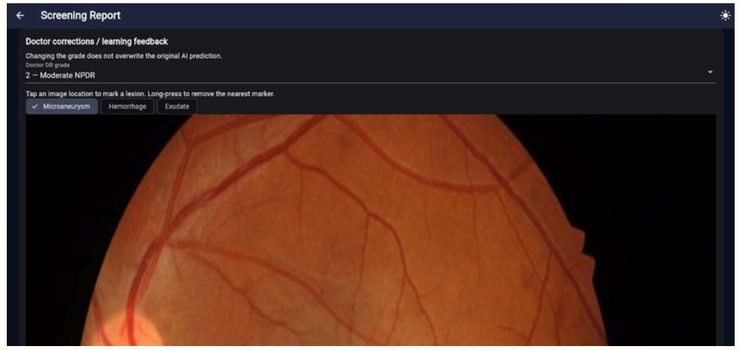
WORKING DEMO AND RESULTS

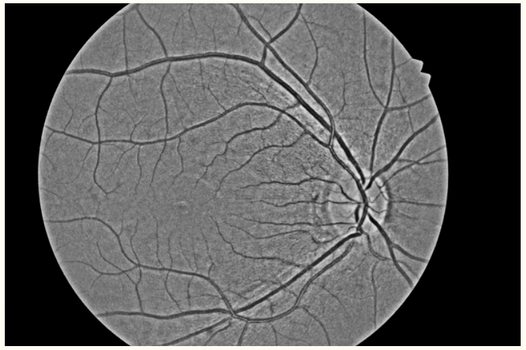
OPTIQ

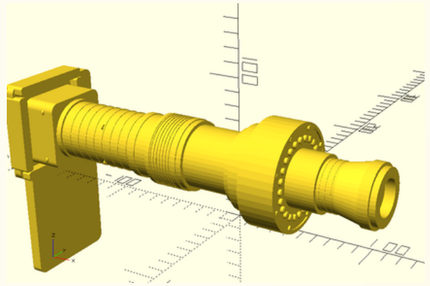


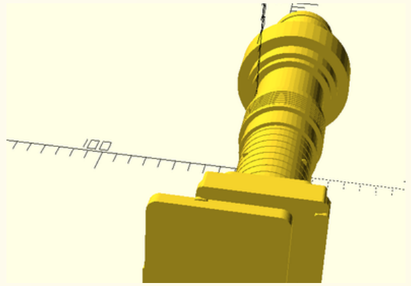




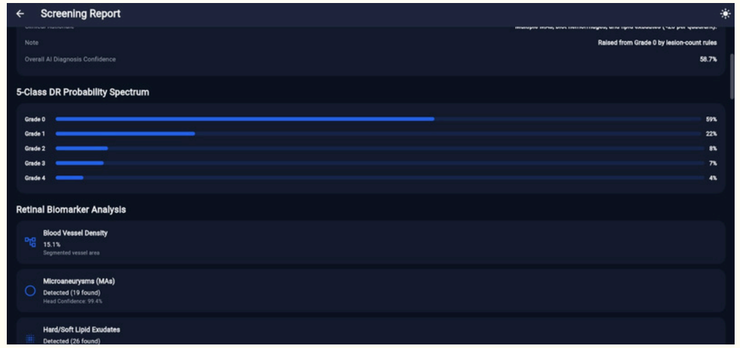








FOVScore	FocusScore	MeanBrightness	BrightnessStd	DarkFraction	BrightFraction








```
------------------------------------------------------------
STARTING EFFICIENTNET-B3 MULTI-TASK TRAINING (364 TRAIN | 91 VAL
------------------------------------------------------------
[Epoch 01] Train Loss: 2.3268 | Val QWK: 0.2843
[Epoch 02] Train Loss: 2.0239 | Val QWK: 0.7438
[Epoch 03] Train Loss: 2.0019 | Val QWK: 0.7821
[Epoch 04] Train Loss: 1.9217 | Val QWK: 0.7174
[Epoch 05] Train Loss: 1.8656 | Val QWK: 0.7687
[Epoch 06] Train Loss: 1.8337 | Val QWK: 0.7438
[Epoch 07] Train Loss: 1.7097 | Val QWK: 0.7119
[Epoch 08] Train Loss: 1.7282 | Val QWK: 0.7376
[Epoch 09] Train Loss: 1.6784 | Val QWK: 0.7429
[Epoch 10] Train Loss: 1.6756 | Val QWK: 0.6882
```

HOW WE ARE AUTHENTIC

INGENUITY AND TECHNICAL DECISIONS

OPTIQ



ORIGINAL AND THOUGHTFUL APPROACH

Combining MATLAB quality filtering with online clinician Reinforcement-Learning-Through-Human-Feedback (RLHF) to catch bad scans upfront while continuously learning from expert feedback.



DIFFERENCE FROM EXISTING SOLUTIONS

Unlike static black-box tools (Google ARDA/IDx-DR), NetraSense pairs 3D-printed affordability with explainable lesion masks and adaptive weight tuning.



IMPORTANT TECHNICAL DECISION

Implementing online clinician RLHF via spatial policy updates so doctors can correct missed lesions or false positives directly on scans.



CHALLENGE AND SOLUTION

Challenge: RLHF updates caused catastrophic forgetting across datasets.

Solution: Added KL-divergence anchoring against a frozen reference model.

BEYOND THE HACKATHON

POTENTIAL AND NEXT STEPS

OPTIQ

WHO COULD USE IT BEYOND THE HACKATHON?

- PHCs and rural clinics with limited retinal imaging
- Diabetes clinics and hospitals for preliminary screening
- NGOs, community eye camps and teleophthalmology networks

WHAT VALUE DO WE PROVIDE?

- 3D-printed smartphone design cuts hardware costs
- AI quality checks speed up screening and triage
- Modular setup and digital reports expand access

WHAT WOULD WE IMPROVE OR BUILD NEXT?

- Refine optics, illumination, alignment and distance sensing
- Explainable AI: Lesion Detection
- Test image quality across lighting and users; add secure clinician feedback

WHAT NEEDS TO HAPPEN FOR REAL-WORLD USE?

- Validate accuracy and rejection rates in clinical trials
- Test prospectively across diverse populations and devices
- Establish regulatory oversight, training, upkeep, and referral pathways

GOAL: AFFORDABLE SCREENING. RELIABLE EVIDENCE. APPROPRIATE REFERRALS.

THANK YOU